# Module - II

## # Basic Elements of an Automated System

An automated system involves the use of technology, software and hardware to perform tasks without human intervention. It aims to improve efficiency, accuracy and consistency while reducing the need for manual labor.

The key elements are:

1. **Input:**

This refers to the data, instructions or raw materials fed into the system for processing. Inputs can be in the form of physical objects (like raw materials in a manufacturing plant) or digital data (like customer orders in an e-commerce system).

For example; In a manufacturing system, the input might be raw materials like metal sheets or plastic. In a software system, the input might be a user query or a set of data for analysis.

2. **Process:**

The process is the set of operations the system performs on the input to convert it into a desired output. The processing might involve calculations, sorting, assembly or transformation of data.

For example; In a robotic arm, the process could involve cutting or welding metal sheets to assemble a car part.

3. Output:

Output is the result of the processing. It can be in the form of physical products, reports, decisions or digital data sent to another system.

For example; In manufacturing, the output might be a finished product, like a car engine or smartphone.

4. Feedback:

Feedback is the information that helps the system to self-correct and improve its operation. This element ensures that the system is adaptive and continuously optimized.

For example; In an automated thermostat, the feedback would be the temperature reading from the sensor, which adjusts the heating or cooling to maintain the desired temperature.

5. Control:

Control refers to the system's ability to direct and regulate the processes using predefined rules, sensors and actuators. It typically uses logic or algorithms to make decisions based on input data, feedback and desired outcomes.

For example; In automated production lines, the control system ensures that each part of the production process runs smoothly and in the correct order, making real-time adjustments as needed.

6. Communication:

Communication refers to the flow of information between different parts of the automated system. This may include sensors, controllers, actuators or external systems. It can be wired or wireless.

For example; In smart factories, communication happens between sensors, robots and the central control system to ensure synchronization in production.

# Control System in Automation

A control system in automation is a set of devices and software that manages, commands, directs or regulates the behavior of other devices or systems. It ensures that a process operates in a desired manner, often without human intervention. Control systems play a vital role in maintaining stability, accuracy, efficiency and safety in automated systems.

## What is a Control System?

A control system monitors the output of a process and compares it to the desired value or setpoint. If there's a difference (called an error), the system takes corrective action through actuators, commands or adjustments.

## Types of Control System

### 1. Open-Loop Control System:

It does not use feedback to determine if the desired output has been achieved. If follows predefined instruction.

For example; A microwave oven running for 2 minutes regardless of whether the food is hot.

## 2. Closed-Loop Control System:

It uses sensors to measure the actual output and adjusts the input accordingly. It ensures better accuracy and adaptability.

For example; a thermostat that runs heating on/off to maintain a set room temperature.

### Components of a Control System

- Sensor - measures physical conditions (e.g. temperature, pressure, position).
- Controller - brain of the system, compares actual output with the desired value.
- Actuator - Converts the control signal into physical action (e.g. motor, valve).
- Setpoint - desired value or condition.
- Process/Plant - system being controlled.
- Feedback Loop - signal sent back from the output to influence the input.

# Examples of Control System in Automation

1. **HVAC Systems (Heating, Ventilation & Air Conditioning):**
   - **Sensor:** Measures room temperature.
   - **Controller:** Compares with set temperature.
   - **Actuator:** Turns heating/cooling on or off.
2. **Industrial Robots:**
   - **Sensor:** Monitors arm position.
   - **Controller:** Adjusts motors to follow programmed path.
   - **Actuator:** Moves robot arm joints accordingly.
3. **Conveyor Belt System:**
   - **Sensor:** Detects object presence.
   - **Controller:** Starts/stops belt or routes object.
   - **Actuator:** Motors control movement or diverters.

# Case Study: Control System in a Car Assembly Plant (Toyota)

**Context:** Toyota uses automated robotic arms to weld, paint and assemble vehicles with minimal human input. Precision and timing are critical to maintain quality and safety.

**Control System Used:** Closed-loop feedback control system with Programmable Logic Controllers (PLCs) and robotic controllers.

**How It Works:** The system automates production through four integrated stages using sensors and dedicated controllers to ensure precision and quality.

- **Welding Arm** (Sensors & PLC) - adjusts power and position in real-time for precise welds.
- **Paint Shop** (Sensors & AI Controller) - monitors coverage and adjusts spray patterns.
- **Coveyor Line** (Proximity Sensors & PLC) - detects chassis to synchronize robotic arms.
- **Quality Inspection** (Vision System & Feedback Controller) - captures images, detects defects, sends signals to halt or correct process.

## Results:

- Improved production efficiency.
- Enhanced product quality.
- Reduced human error.
- Real-time fault detection and correction.

## Advantages of Control Systems in Automation

- **Precision**: Keeps the system operating at desired parameters.
- **Reliability**: Reduced variability and downtime.
- **Adaptability**: Adjusts in real-time to changing conditions.
- **Safety**: Automatically responds to abnormal situations.
- **Efficiency**: Saves energy and material by minimizing waste.

## Real-World Applications of Control Systems

- Automotive (Cruise control, abs braking, airbag deployment)
- Manufacturing (CNC machines, process control)
- Aerospace (Flight control systems, autopilots)
- Smart Homes (Automated lighting, security, climate control)
- Power Plants (Load balancing, turbine control)

# Advanced Automation Functions

Advanced Automation Function refer to complex, sophisticated processes designed to automate tasks that would typically require significant human intervention. These functions go beyond simple task automation to incorporate more intricate systems like decision-making, machine learning, predictive analytics and advanced data processing.

## Key Components of Advanced Automation

1. **Decision-making Automation:**
   Systems dynamically choosing actions from live data or historical trends (finance, healthcare).

2. **Robotic Process Automation (RPA):**
   Software bots automating rule-based business tasks, often enhanced with ML.

3. **AI Integration:**
   Machine learning and NLP handling complex customer interactions and efficiency predictions.

4. **IoT & Smart Systems:**
   Networked physical devices streaming operational data to trigger real-time adaptations.

5. Predictive Maintenance:
Analyzing machine run metrices to fix problems before breakdowns occur.

6. Self-Learning & Adaptation:
Algorithms automatically refining parameters over time based on past performance.

## Examples of Advanced Automation Functions

1. Automated Data Analysis with AI & ML:
Real-time tailoring of e-commerce recommendations to individual customer habits.

2. Self-Healing IT Systems:
Cloud systems diagnosing degradation and automatically restarting services or scaling resources.

3. Advanced Manufacturing Automation:
Vision-inspected, self-adjusting robotic production lines.

4. Automated Customer Support:
Chatbots and Virtual Assistants - NLP agents resolving queries, understanding intent and learning from interactions.

5. Supply Chain Optimization:
AI forecasting demand, scheduling stock replenishment and re-routing logistics.

## Case Study: Automated Warehouse Management at Amazon

**Problem**: Scaling high-volume fulfillment while eliminating seasonal bottlenecks and manual errors.

**Solution**: Combined RPA, IoT & AI.

1. **Kiva Robots** - navigate autonomously to haul pods to human packers.
2. **AI-Powered Inventory** - Optimizes item placement near packers based on demand forecasts.
3. **Predictive Maintenance** - IoT sensors detect component wear on conveyors/robots to schedule repairs.
4. **Order Fulfillment Optimization** - Pathfinding models route robots along the fastest paths.

**Outcome**: Order processing dropped from hours to minutes, reduced labor overhead per order and drastically lower packing errors.

# Levels of Automation

Levels of Automation (LOA) define the proportion of operational control and decision-making transferred from human operators to machines.

## What are Levels of Automation?

A spectrum tracing assistance from purely manual work to complete autonomous execution.

## Five Levels of Automation (Industrial Perspective)

Level-0 (Manual Operations): Human performs all tasks manually; no automation.

Level-1 (Assisted Automation): Human performs tasks using assistive machine guidance (e.g. barcode scanners, calculators).

Level-2 (Partial Automation): System performs designated tasks under human supervision and triggering (e.g. CNC machines).

Level-3 (Conditional Automation): System executes most tasks autonomously, human intervenes only during exception (e.g. highway autopilot).

Level-4 (Full Automation): System operates, monitors and self-corrects without human presence (e.g. fully automated warehouses.

# Case Study: Siemens Smart Factory

**Company**: Siemens AG, Amberg Plant (Germany).

**Problem**: Needed higher PLC output and fewer defects without adding staff.

**Solution (Gradual Automation)**:

- **Level 0 to 2** - Transitioned from hand assembly to tool-assisted, semi-automated lines.
- **Level 3** - Integrated sensor feedback networks to assist operator decision-making.
- **Level 4** - Rebuilt into a 75% automated digital factory with machine-to-machine coordination and real-time self-correction.

**Results**: Defect rate dropped from ~1% to <0.001%; switched to 24/7 continuous output; shifted labor entirely to supervisory maintenance.

# Sensors

A sensor detects physical environmental inputs (light, heat, motion, pressure) and converts them into readable or processable electronic signals.

## Components of a Sensor System

1. **Sensing Element**: Directly senses the physical condition.
2. **Transducer**: Converts detected parameters into an electrical signal.
3. **Signal Processing Unit**: Amplifies, filters and digitizes analog signals.
4. **Output Interface**: Transmits processed information to displays or controllers.

## Types of Sensors

1. **Temperature Sensor**: Measures heat/cold (e.g. Thermistor, Thermocouple; HVAC).
2. **Proximity Sensor**: Measures distance (e.g. IR, Ultrasonic; parking sensors, faucets).
3. **Pressure Sensor**: Measures applied force (e.g. Barometer, Piezoresistive; tire monitors).

4. Light Sensor: Measures light intensity (e.g. LDR, Photodiode; auto-lighting).

5. Motion Sensor: Measures physical displacement (e.g. PIR; security alarms).

6. Gas Sensor: Measures airborne gas concentration (e.g. MQ2; air quality monitors).

7. Humidity Sensor: Measures moisture in air (e.g. Hygrometer, DHT11; weather stations).

8. Sound Sensor: Measures sound level/pressure (e.g. Microphone; voice recognition).

9. Touch Sensor: Detects physical contact (e.g. Capacitive; touchscreens).

## Applications of Sensors

- Healthcare - ECGs & Glucose monitors.
- Industrial Automation - Condition monitoring
- Automotive - Airbags
- Smart Homes - detectors
- Environment Monitoring - pollution meters.

"Sensors act as the digital eyes & ears."

# Actuators

An actuator converts input energy (electrical, hydraulic, ~~or~~ pneumatic) into physical mechanical motion - acting as the muscles of an automated setup.

## How Actuators Work

- The control system sends an electrical command.
- The actuator converts the signal into linear or rotary displacement.
- The motion drives a physical plant (opens valves, rotates wheels, articulates joints).

## Types of Actuators

1. Electric Actuators: Converts electricity into linear/rotary motion (e.g. Servo motors, stepper motors).
2. Hydraulic Actuators: Uses high-pressure fluid for heavy linear forces (e.g. Hydraulic cylinders).
3. Pneumatic Autuators: Uses compressed air for fast linear actions (e.g. Pneumatic pistons).

4. Thermal Actuators: Uses heat expansion to generate motion (e.g. Thermostat wax motors).

5. Magnetic Actuators: Uses magnetic flux to shift mechanisms (e.g. Solenoids).

## Case Study: Industrial Robotic Arm

Background: Automating car door installations to reduce cycle times and assembly flaws.

Solution: System incorporating servo motors (arm angles), pneumatic grippers (door handling) and hydraulic actuators (final press-fit) guided by proximity or pressure sensors.

Results: Door install time reduced by 35%, alignment errors eliminated, labor costs minimize.

## Applications of Actuators

- Automotive (power windows, steering)
- Home Automation (smart blinds)
- Medical Devices (ventilators, surgical robots)
- Industrial Automation (conveyors)
- Aerospace (wing flaps, rudders)

# Analog-Digital Conversions (ADC)

ADC is the electronic conversion of continuous analog physical signals (voltage, sound, temperature) into discrete binary numbers compatible with microprocessors.

## Why is ADC Important?

Real-world signals vary continuously, but computers require binary code to compute and store data.

## Steps in ADC

1. Sampling: Measuring continuous signal amplitude at discrete periodic time intervals.
2. Quantization: Rounding sampled continuous voltages into defined finite steps.
3. Encoding: Converting quantized discrete levels into digital binary words.

## Types of ADC

1. Successive Approximation Register (SAR): Employs binary search logic; offers high accuracy at moderate speeds (common in microcontrollers).

2. Flash ADC: Uses a parallel array of comparators; ultra-fast conversion at moderate accuracy (e.g. oscilloscopes).

3. Sigma-Delta: Uses oversampling and noise shaping; very high resolution at slower speeds (e.g. audio systems).

4. Dual Slope: Integrates voltage over time; slow but exceptionally accurate (digital multimeters).

5. Pipelined: Divides conversion across sequential sub-stages for high throughput (e.g. video and imaging).

Case Study: ADC in Electrocardiogram Machine

Background: Heartbeats produce micro-level analog electrical impulses needing digital capture.

Implementation: Electrodes detect heart signals; SAR ADC samples the signal at 250 to 500 Hz; binary data is analyzed for cardiac anomalies.

Outcome: Accurate digital heart rhythm analysis and cloud tele-monitoring.

Applications → Audio gear, medical monitors, etc.

# Input/Output Devices for Discrete Data

## Discrete Input Devices

Binary field switches transmitting ON/OFF or HIGH/LOW binary states to the controller (e.g. pushbutton switches, limit switches, selector switches and optical proximity sensors).

## Discrete Output Devices

Binary actuators energized or de-energized by controller commands (e.g. indicator lights, alarm horns, motor starter contactors and pneumatic solenoid values.

# Contact Input/Output Interfaces

## Contact Input Interfaces

Input conditioning circuits designed to read mechanical switch closures safely; includes signal debouncing, voltage level translation and optoisolators to protect digital CPU boards from industrial voltage spikes.

## Contact Output Interfaces

Switching stages enabling low-power digital controller logic to switch high-power industrial plant equipment.